Overview of Kalman Filter Theory and Applications

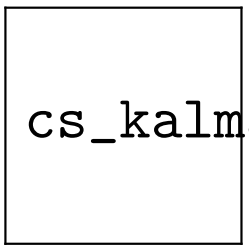
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Topics

- Control Segment Ephemeris and Clock Estimator (ECEST)
- ECEST State Variables
- ECEST Measurement Model
- ECEST Partitioning


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Control Segment Ephemeris and Clock Estimator (ECEST)

- Kalman Filter to estimate satellite ephemeris residuals and clock errors
  - Inertial positions and velocities of SV
  - SV and Monitoring Station (MS) clock biases and their derivatives
  - Solar pressure parameters
  - Tropospheric state at each MS

Control Segment Ephemeris and Clock Estimator (ECEST)

- Hundreds of state variables
  - Primary variables of interest are SV position, velocity and clock errors, because these are used directly by the the User Equipment to compute user position, velocity and time.
- Partitioned into decoupled blocks based on SV-MS pairs

Residual States vs. Whole States

- Epoch State Residual
- Reference Trajectory for each SV
- Residual is difference between reference trajectory and the true state
- Residual states sometimes called error states

ECEST State Variables – Position

- $r_s(t)$ = current position vector of the center of gravity of the SV body in Earth-centered, inertial coordinates (meters).
- $r_{s,ref}(t)$ = nominal (reference) position vector of the center of gravity of the SV body in Earth-centered, inertial coordinates (meters)
- $r_{sa}(t)$ = current position vector of SV antenna phase center in Earth-centered, inertial coordinates (meters)
- $r_m(t)$ = current position vector of MS antenna in Earth-centered, inertial coordinates (meters)

ECEST State Variables – SV and MS Antenna States

- $r_{\mathrm{sa}}(t)$ = current position vector of SV antenna phase center in Earth-centered, inertial coordinates (meters)
- $r_{\mathrm{m}}(t)$ = current position vector of MS antenna in Earth-centered, inertial coordinates (meters)
- $r_{\mathrm{ma}}(t)$ = current position vector of the MS antenna phase center in Earth-centered, inertial coordinates (meters)

ECEST State Variables – Velocity States

- $v_s(t)$ = current velocity vector of the center of gravity of the SV body in Earth-centered, inertial coordinates (meters/sec)
- $v_{s,ref}(t)$ = nominal (reference) velocity vector of the center of gravity of the SV body in Earthcentered, inertial coordinates (meters/sec)

ECEST State Variables – Current Solar Pressure Parameters

- $k_1(t)$ = current solar pressure scaling parameter along the sunline (unitless)
- $k_2(t)$ = current solar pressure parameter for unmodeled acceleration in force model (meters/sec^2)

Bundled as mini-state vector:

$$\mathrm{KR}(t) = \begin{bmatrix} k_1(t) \\ k_2(t) \end{bmatrix}$$

ECEST State Variables – Nominal (reference) Solar Pressure Parameters

- $k_{1,\text{ref}}(t)$ = nominal (reference) solar pressure scaling parameter along the sunline (unitless)
- $k_{2,\text{ref}}(t)$ = nominal (reference) solar pressure parameter for unmodeled acceleration in force model (meters/sec^2)

Also bundled as mini-state vector:

$$\text{KR}_{\text{ref}}(t) = \begin{bmatrix} k_{1,\text{ref}}(t) \\ k_{2,\text{ref}}(t) \end{bmatrix}$$

ECEST State Variables – Current SV Clock States

- $\tau_s(t)$ = current SV clock bias (sec)
- $\dot{\tau}_s(t)$ = current SV clock drift (sec/sec)
- $\ddot{\tau}_s(t)$ = current SV clock drift rate (sec/sec^2)

Bundled as a mini-state vector:

$$\tau_s(t) = \begin{bmatrix} \tau_s(t) \\ \dot{\tau}_s(t) \\ \ddot{\tau}_s(t) \end{bmatrix}$$

ECEST State Variables – Nominal (reference) SV Clock States

- $\tau_{\mathrm{s,ref}}(t)$ = nominal (reference) SV clock bias (sec)
- $\dot{\tau}_{\mathrm{s,ref}}(t)$ = nominal (reference) SV clock drift (sec/sec)
- $\ddot{\tau}_{\mathrm{s,ref}}(t)$ = nominal (reference) SV clock drift rate (sec/sec^2)

Also bundled as a mini-state vector:

$$\tau_{\mathrm{s,ref}}(t) = \begin{bmatrix} \tau_{\mathrm{s,ref}}(t) \\ \dot{\tau}_{\mathrm{s,ref}}(t) \\ \ddot{\tau}_{\mathrm{s,ref}}(t) \end{bmatrix}$$

ECEST State Variables – MS Clock States

- $\tau_m(t)$ = current MS clock bias (sec)
- $\dot{\tau}_m(t)$ = current MS clock drift (sec/sec)
- $\tau_{m,\text{ref}}(t)$ = nominal (reference) MS clock bias (sec)
- $\dot{\tau}_{m,\text{ref}}(t)$ = nominal (reference) MS clock drift (sec/sec)

ECEST State Variables – Tropospheric States

- $X_T(t)$ = current wet tropospheric radial height relative to the sensor (meters)
- $X_{T,ref}(t)$ = nominal (reference) wet tropospheric radial height relative to the sensor (meters)

ECEST State Vector Model

- For each SV there are 11 state variables

$$[r_s, v_s, \mathrm{KR}, \tau_s]^\top$$

- For each MS there are 3 state variables:

$$[\tau_m, X_T]^\top$$

- Thus, with 30 SVs and 20 MSs we have

$$30 \times 11 + 20 \times 3 = 390 \text{ possible states.}$$

ECEST Measurement Model

- Observations / Measurement model provided by Monitoring Stations (MSs)
- smoothed, corrected pseudorange (SPR_C)
- corrected, accumulated delta range (ΔADR_C).
- corrections are made at MCS by removal of ionospheric delay

ECEST Measurement Model

Each MS sends via UDP:

- smoothed 1.5-sec L1/L2 Pseudorange and Accumulated Delta Range every 15 min (Msg 83)
- local weather measurements every 15 min (Msg 81)

Each MS also sends via UDP (but not usesd in Kalman filter):

- four 1.5-sec raw L1/L2 Pseudorange and Accumulated Delta Range measurements every 6 sec (Msg 80)

ECEST Measurement Model

- Each SV-MS observation pair yields two measurements

$$[\mathrm{SPR}_C,\ \Delta\mathrm{ADR}_C]^\top$$

- Each SV can be paired with as many as 10 MSs
- Thus, observation space can also be large (dim ≤ 30)

ECEST Partitioning

- To reduce the size of the state and observation spaces, a block matrix partitioning scheme is employed.
  - Each partition contains up to all SVs and always contains all MS States
  - Each SV belongs to one and only one partition at any instant of time.
  - Each partition esimtation is idnependent of other partitions.